

5. Shekhar, S., and Carlier, M.F. (2017). Enhanced depolymerization of actin filaments by ADF/cofilin and monomer funneling by capping protein cooperate to accelerate barbed-end growth. *Curr. Biol.* 27, 1990–1998.e5.
6. Wioland, H., Guichard, B., Senju, Y., Myram, S., Lappalainen, P., Jegou, A., and Romet-Lemonne, G. (2017). ADF/cofilin accelerates actin dynamics by severing filaments and promoting their depolymerization at both ends. *Curr. Biol.* 27, 1956–1967.e7.
7. Kotila, T., Wioland, H., Enkavi, G., Kogan, K., Vattulainen, I., Jegou, A., Romet-Lemonne, G., and Lappalainen, P. (2019). Mechanism of synergistic actin filament pointed end depolymerization by cyclase-associated protein and cofilin. *Nat. Commun.* 10, 5320.
8. Shekhar, S., Chung, J., Kondev, J., Gelles, J., and Goode, B.L. (2019). Synergy between Cyclase-associated protein and Cofilin accelerates actin filament depolymerization by two orders of magnitude. *Nat. Commun.* 10, 5319.
9. Carlier, M.F., Laurent, V., Santolini, J., Melki, R., Didry, D., Xia, G.X., Hong, Y., Chua, N.H., and Pantaloni, D. (1997). Actin depolymerizing factor (ADF/cofilin) enhances the rate of filament turnover: implication in actin-based motility. *J. Cell Biol.* 136, 1307–1322.
10. Towsif, E.M., and Shekhar, S. (2025). The actin filament pointed-end depolymerase Srv2/CAP depolymerizes barbed ends, displaces capping protein, and promotes formin processivity. *Proc. Natl. Acad. Sci. USA* 122, e2411318122.
11. Kotila, T., Kogan, K., Enkavi, G., Guo, S., Vattulainen, I., Goode, B.L., and Lappalainen, P. (2018). Structural basis of actin monomer re-charging by cyclase-associated protein. *Nat. Commun.* 9, 1892.
12. Ghosh, M., Song, X., Mouneimne, G., Sidani, M., Lawrence, D.S., and Condeelis, J.S. (2004). Cofilin promotes actin polymerization and defines the direction of cell motility. *Science* 304, 743–746.
13. Chaudhry, F., Breitsprecher, D., Little, K., Sharov, G., Sokolova, O., and Goode, B.L. (2013). Srv2/cyclase-associated protein forms hexameric shurikens that directly catalyze actin filament severing by cofilin. *Mol. Biol. Cell* 24, 31–41.
14. Normoyle, K.P., and Brieher, W.M. (2012). Cyclase-associated protein (CAP) acts directly on F-actin to accelerate cofilin-mediated actin severing across the range of physiological pH. *J. Biol. Chem.* 287, 35722–35732.
15. Nguyen, P.D.N., Abe, H., Umeda, K., Ono, S., and Kodera, N. (2026). Actin monomers influence the interaction between *Xenopus* cyclase-associated protein 1 and actin filaments. *J. Mol. Biol.* 438, 169796. <https://doi.org/10.1016/j.jmb.2026.169796>.
16. Kudryashova, E., Ankita, Ulrichs, H., Shekhar, S., and Kudryashov, D.S. (2022). Pointed-end processive elongation of actin filaments by *Vibrio* effectors VopF and VopL. *Sci. Adv.* 8, eadc9239.
17. Shekhar, S., Fowler, V.M., and Gregorio, C.C. (2026). Renaissance at the actin filament pointed end: Mechanisms of assembly, capping and depolymerization. *Curr. Opin. Cell Biol.* 98, 102602.
18. Bombardier, J.P., Eskin, J.A., Jaiswal, R., Correa, I.R., Jr., Xu, M.Q., Goode, B.L., and Gelles, J. (2015). Single-molecule visualization of a formin-capping protein ‘decision complex’ at the actin filament barbed end. *Nat. Commun.* 6, 8707.
19. Shekhar, S., Kerleau, M., Kuhn, S., Pernier, J., Romet-Lemonne, G., Jegou, A., and Carlier, M.F. (2015). Formin and capping protein together embrace the actin filament in a ménage à trois. *Nat. Commun.* 6, 8730.
20. Ulrichs, H., Gaska, I., and Shekhar, S. (2023). Multicomponent regulation of actin barbed end assembly by twinfilin, formin and capping protein. *Nat. Commun.* 14, 3981.

Neuroethology: Seeing to steer in flies

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<https://doi.org/10.1016/j.cub.2026.04.043>

Blowfly motion vision is tuned not simply to represent optic flow, but to support rapid flight control. A recent study shows that blowfly visual neurons preferentially encode the self-motion signals most useful for stabilisation, revealing an embodied principle of sensory coding.

Why does a fly see the world the way it does: to encode the environment efficiently, or to control its body effectively? For decades, an influential answer in sensory neuroscience came from efficient coding: the idea that sensory systems are adapted to the statistical structure of natural inputs and use limited neural resources economically^{1,2}. In vision, this has often been taken to imply that neural coding should reduce redundancy and allocate limited resources to the most informative features of the visual world. But for a flying insect that must remain upright, stable and manoeuvrable while being

continually buffeted by gusts and reshaped by its own motor commands, efficient coding of the visual world may not be the main goal. Under those conditions, vision matters not simply because it describes the world, but because it helps stabilise an unstable body. Humbert *et al.*³ have now given that intuition a formal control-theoretic basis, arguing that blowfly motion vision is tuned less to encoding optic flow in general, and more to supporting the control of that unstable body in flight.

This argument extends, rather than overturns, a long neuroethological

tradition in which fly lobula plate tangential cells were first described as decoders of retinal image flow⁴, then shown to encode specific combinations of self-motion^{5,6}, and later formalised as matched filters for optic flow⁷. Elementary motion detectors earlier in the pathway are likewise adapted to each species’ visual ecology^{8,9}. Later work under naturalistic conditions showed that motion-sensitive responses are shaped by the statistics of free flight, by adaptation, by behavioural state, and by visuomotor demands^{10–14}. Together, these studies established that fly motion vision is highly specialised for



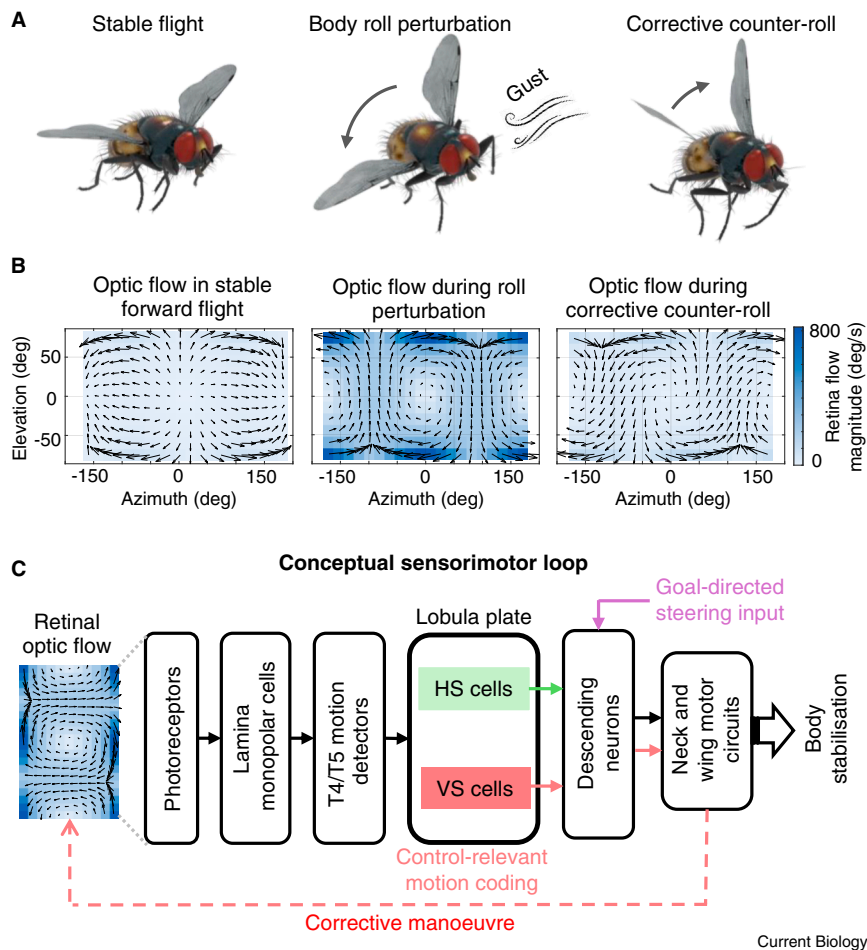


Figure 1. Motion vision supports flight stabilisation through an embodied sensorimotor loop.

(A) A blowfly in stable forward flight is displaced by a gust that, in this example, induces a body roll perturbation. The perturbation alters body orientation and trajectory, eliciting compensatory responses that restore stability. (B) Corresponding retinal optic-flow fields for each flight state. Stable forward flight generates a characteristic wide-field flow pattern, whereas a wind-driven roll perturbation introduces a stronger rotational component across the retina. During corrective stabilisation, this perturbation-associated optic flow is reduced as compensatory action counteracts the roll. (C) Conceptual summary of the sensorimotor loop linking visual motion to stabilising action. Retinal optic flow is processed sequentially by photoreceptors, lamina monopolar cells and T4/T5 motion detectors before converging onto lobula plate circuits, including HS and VS cells, which encode control-relevant self-motion signals. These signals are relayed to descending neurons and then to neck and wing motor circuits, whose output contributes to body stabilisation. The dashed red pathway indicates the negative-feedback logic of the loop: corrective manoeuvres alter body motion, which in turn reshapes the retinal optic flow driving the visual pathway. The magenta pathway denotes an additional goal-directed steering input to descending neurons, highlighting how visually guided stabilisation can interact with internal steering commands. Together, the figure illustrates the idea that motion vision is tuned not simply to represent optic flow but to emphasise the self-motion signals most useful for rapid stabilisation and steering.

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particularly appropriate because flight poses exactly the kind of problem that control theory was developed to address: how to maintain stability in a system continually challenged by perturbations, delays and changing conditions, and to do so through fast feedback. The authors combine comprehensive electrophysiological recordings from identified motion-sensitive neurons in the blowfly *Calliphora* with measurements of body form, free-flight kinematics and aerodynamic modelling to build a dynamical model of blowfly flight that links mechanical input, body dynamics and neural output over time. In this framework, steering commands and aerodynamic perturbations move the body, those movements generate characteristic patterns of optic flow, and identified visual neurons encode the resulting self-motion in ways that can guide rapid corrective motor output, thereby feeding back into body dynamics (Figure 1A,B). This is what makes the study distinctive: rather than analysing receptive fields as isolated visual filters, the authors place them within the full dynamical chain that connects perturbation, body motion, sensory signalling and corrective control.

This reframing is important because, from the perspective of flight control, not all body motions matter in the same way. Some are easier for the fly to generate, some are more strongly driven by aerodynamic perturbations, and some leave a clearer signature in the responses of visual neurons. Humbert *et al.*³ analyse these differences using three related control-theoretic measures: controllability, which captures how readily a motion can be produced by the fly; observability, which captures how clearly that motion is reflected in neural output; and disturbance sensitivity, which captures how strongly it is driven by external perturbations. Within the motion pathway, the vertical-system and heterolateral V cells, both classes of lobula plate tangential cell, are associated mainly with combinations of roll, pitch and heave important for flight stabilisation. Their tuning is therefore not explained simply by which motions are most clearly reflected in visual output, but is better understood in relation to the self-motion components

self-motion during flight and is better understood not as an isolated sensory code, but as part of a closed sensorimotor loop. What remained unresolved was why these particular motion patterns are privileged: are they tuned to represent optic flow as a visual pattern, to estimate self-motion, or to

support the control of an unstable flying body? Humbert *et al.*³ now provide a more explicit framework for answering that question³.

In a technical *tour de force*, Humbert *et al.*³ bring sensory physiology, flight mechanics and control theory into a single framework. That framework is

most relevant for stabilising the body in flight.

The deeper result emerges when Humbert *et al.*³ ask which directions of self-motion matter most for both sensing and control. To address this, they use open-loop Hankel singular values, a control-theoretic measure of how effectively signal energy, understood here as the strength of behaviourally relevant motion signals, is transferred from aerodynamic disturbances and control inputs to sensor outputs. This shifts the argument from separate comparisons to an integrated one: not which motions are most controllable, most observable, or most disturbance-sensitive in isolation, but which motions matter most when the sensorimotor system is considered as a whole. By this view, the vertical-system and heterolateral V cells appear especially well suited to stabilisation and disturbance rejection, whereas the horizontal-system and H cells align more strongly with forward motion, yaw and sideslip, suggesting roles more closely related to guidance and navigation than to rapid attitude correction (Figure 1C). The study can thus be read as a refinement of the earlier mode-sensing hypothesis¹⁴: the motion pathway does appear to be organised around dynamically important modes of self-motion, but Humbert *et al.*³ show that the deeper principle is more precise, namely that receptive fields are tuned to maximise the transfer of behaviourally useful signal energy through the sensorimotor loop. The point, then, is not that some neurons ‘see’ while others ‘control’: all are visual neurons; what differs is the role they play within the sensorimotor loop linking sensation, body dynamics and action (Figure 1).

From this perspective, the Humbert *et al.*³ study reframes much of the older literature. The matched-filter idea survives, but its meaning becomes more precise. These neurons are not just matched to optic flow as a geometrical pattern; they are matched to the consequences of living in a particular moving body. This interpretation is also supported by more recent work in *Drosophila*, which has begun to reveal how self-motion signals are routed and used within sensorimotor circuits. Descending visual interneurons carry self-motion signals towards the motor

centres¹⁵; optic-flow processing contributes directly to course control¹⁶; and recent circuit studies have shown how bilateral interactions and disinhibitory motifs help extract behaviourally useful motion signals under natural conditions^{17,18}. Humbert *et al.*³ add a complementary level of explanation. They do not identify the full circuit mechanism, but they do offer a principle for why such circuits should privilege some self-motion directions over others.

The Humbert *et al.*³ paper also raises important questions. Its analysis is necessarily linearised around a particular operating point, whereas real flight is nonlinear and unfolds in a continuously closed loop¹⁹. It will therefore be important to test whether the same control-relevant motion patterns identified here can be traced through descending neurons, steering muscles and behaviour in freely flying animals. Recent work in *Drosophila* has shown that specific steering muscles can embody distinct elements of a stabilising controller²⁰, suggesting a plausible route from visual tuning to motor implementation.

Overall, the conceptual advance is already clear. Humbert *et al.*³ argue that, in the blowfly, motion vision is efficient not because it provides a general description of optic flow, but because it gives priority to the sensory consequences of the body motions that matter most for control³. This is why this work reaches beyond insect flight. It suggests that sensory systems may be shaped not only by the statistics of the external world, but also by the control demands of the body that must act within the world. More broadly, the study suggests that evolution may tune sensors jointly for sensing and control, not for representation alone. For neuroscience, this offers a more embodied view of sensory coding. For bio-inspired machines, it points to a different design principle: sensors may work best not when they reconstruct the world in full, but when they selectively emphasise the signals most useful for stabilisation, steering and rapid action.

DECLARATION OF INTERESTS

The author declares no competing interests.

REFERENCES

1. Barlow, H.B. (1961). Possible principles underlying the transformation of sensory messages. In *Sensory Communication*, W.A. Rosenblith, ed. (MIT Press), pp. 217–233.
2. Laughlin, S.B., van Steveninck, R.R.D., and Anderson, J.C. (1998). The metabolic cost of neural information. *Nat. Neurosci.* **1**, 36–41.
3. Humbert, J.S., Krapp, H.G., Baeder, J.D., Badrya, C., Dawson, I.L., Huang, J.V., Hyslop, A., Jung, Y.S., Leroy, A., Lutkus, C., *et al.* (2026). Fly motion vision maximizes signal energy transfer between mechanical input and sensor output. *Sci. Robot.* **11**, eadx7524.
4. Hausen, K. (1993). Decoding of retinal image flow in insects. *Rev. Oculomot. Res.* **5**, 203–235.
5. Krapp, H.G., Hengstenberg, B., and Hengstenberg, R. (1998). Dendritic structure and receptive-field organization of optic flow processing interneurons in the fly. *J. Neurophysiol.* **79**, 1902–1917.
6. Krapp, H.G., and Hengstenberg, R. (1996). Estimation of self-motion by optic flow processing in single visual interneurons. *Nature* **384**, 463–466.
7. Franz, M.O., and Krapp, H.G. (2000). Wide-field, motion-sensitive neurons and matched filters for optic flow fields. *Biol. Cybernet.* **83**, 185–197.
8. O’Carroll, D., Bidwell, N., Laughlin, S., and Warrant, E. (1996). Insect motion detectors matched to visual ecology. *Nature* **382**, 63–66.
9. O’Carroll, D.C., Barnett, P.D., and Nordström, K. (2011). Local and global responses of insect motion detectors to the spatial structure of natural scenes. *J. Vis.* **11**, 20.
10. Lindemann, J.P., Kern, R., Van Hateren, J., Ritter, H., and Egelhaaf, M. (2005). On the computations analyzing natural optic flow: quantitative model analysis of the blowfly motion vision pathway. *J. Neurosci.* **25**, 6435–6448.
11. Karmeier, K., Van Hateren, J., Kern, R., and Egelhaaf, M. (2006). Encoding of naturalistic optic flow by a population of blowfly motion-sensitive neurons. *J. Neurophysiol.* **96**, 1602–1614.
12. Huston, S.J., and Krapp, H.G. (2008). Visuomotor transformation in the fly gaze stabilization system. *PLoS Biol.* **6**, e173.
13. Maimon, G., Straw, A.D., and Dickinson, M.H. (2010). Active flight increases the gain of visual motion processing in *Drosophila*. *Nat. Neurosci.* **13**, 393–399.
14. Hyslop, A., Krapp, H.G., and Humbert, J.S. (2010). Control theoretic interpretation of directional motion preferences in optic flow processing interneurons. *Biol. Cybernet.* **103**, 353–364.
15. Suver, M.P., Huda, A., Iwasaki, N., Safarik, S., and Dickinson, M.H. (2016). An array of descending visual interneurons encoding self-motion in *Drosophila*. *J. Neurosci.* **36**, 11768–11780.

16. Pokusaeva, V.O., Satapathy, R., Symonova, O., and Joesch, M. (2024). Bilateral interactions of optic-flow sensitive neurons coordinate course control in flies. *Nat. Commun.* 15, 8830.
17. Zhao, A., Gruntman, E., Nern, A., Iyer, N., Rogers, E.M., Koskela, S., Siwanowicz, I., Dreher, M., Flynn, M.A., and Laughland, C. (2025). Eye structure shapes neuron function in *Drosophila* motion vision. *Nature* 646, 135–142.
18. Erginkaya, M., Cruz, T., Brotas, M., Marques, A., Steck, K., Nern, A., Torrão, F., Varela, N., Bock, D.D., and Reiser, M. (2025). A competitive disinhibitory network for robust optic flow processing in *Drosophila*. *Nat. Neurosci.* 28, 1241–1255.
19. Sun, M. (2014). Insect flight dynamics: stability and control. *Rev. Mod. Phys.* 86, 615–646.
20. Whitehead, S.C., Leone, S., Lindsay, T., Meiselman, M.R., Cowan, N.J., Dickinson, M. H., Yapici, N., Stern, D.L., Shirangi, T., and Cohen, I. (2022). Neuromuscular embodiment of feedback control elements in *Drosophila* flight. *Sci. Adv.* 8, eabo7461.

Bacterial defence systems: A lucky charm that starves viral replication

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<https://doi.org/10.1016/j.cub.2026.04.072>

A newly characterized bacterial antiphage system is held inactive by an inhibitory nucleotide signal and activated by infection-driven dTTP accumulation, uncovering a metabolite-responsive, toxin–antitoxin-like defence mechanism that restricts phage replication through selective depletion of the cellular dGTP pool.

The evolutionary arms race between bacteria and their viruses (bacteriophages) has fuelled the proliferation of diverse antiphage defence (immune) systems. Many of these defence systems are activated by sensing phage-induced infection cues, such as phage-specific nucleic acids or proteins. Once activated, they deploy a range of immune response mechanisms that can clear or stall infections, in some cases through effector-induced cellular dormancy. Among these antiviral mechanisms, nucleotide-depleting enzymes known as deoxynucleoside triphosphohydrolases (dNTPases) have been shown to confer protection by depleting the cellular dNTP pool, which is required for viral replication^{1,2}. However, a key question has puzzled researchers: how are defensive dNTPases specifically activated during infection? Recent work by Yamaguchi and colleagues provides an answer to this for Clover — a new class of two-component nucleotide-depletion systems, named after the distinctive shape of their dNTPase effector complexes³.

Clover is just one of a vast arsenal of antiphage defence systems encoded by

bacteria, many of which have only recently been discovered and remain unstudied. With discovery outpacing mechanistic characterisation, important questions remain about many antiphage defence systems, including how they are activated during infection whilst avoiding harmful autoimmune activity in uninfected cells. Clover was previously reported as an unnamed, predicted two-gene nucleotide-targeting defence system in a genomic ‘treasure hunt’ that uncovered multiple new families of putative bacterial immune mechanisms⁴. Yamaguchi *et al.* now show that Clover is a metabolite-responsive phage-defence system, offering a mechanistic explanation for how nucleotide-depleting immunity can be selectively engaged during infection³. They identified more than 650 examples of Clover systems, which are comprised of a dNTPase (CloA) paired with a nucleotidyltransferase (dNTase, CloB), in a wide range of bacteria, including proteobacteria and *Bacillus* species.

Dual nucleotide control of Clover activity

Focusing on representative Clover systems from *Salmonella enterica* and

Escherichia coli, phage-challenge assays were used to demonstrate that both CloA and CloB are required to provide strong protection against T5-like *Demereciviridae* phages. Active-site mutagenesis showed that the dGTPase activity of CloA was essential for phage defence and that CloB activity suppressed CloA-mediated cytotoxicity. The interplay between CloA and CloB aligns with toxin–antitoxin-like interactions, wherein the toxic component (CloA) induces cellular dormancy or death to suppress phage spread, as observed for Clover. Interestingly, CloA was activated by binding the nucleotide deoxythymidine triphosphate (dTTP) and was inhibited by a nucleotide analogue, 5'-triphosphothymidyl-3'-thymidine (p3diT), which is produced by CloB. Cryo-EM structures of the CloA octamer with its dGTP substrate and either dTTP or p3diT show that these molecules bind to distinct allosteric sites (Figure 1). Importantly, CloA was revealed to have a greater affinity for p3diT compared to dTTP, and the p3diT-mediated CloA inhibition could only be overcome by high dTTP levels. Elevation of host dNTPs, such as dTTP, can occur during infection